



National Forensics Sciences University, Goa Campus Mid- semester Examination

Programme- M. Sc. Digital Forensics & Information Security Date- 18-03-2026 Subject Name: Malware Analysis and Forensics Subject Code- CTMSDFIS SII P3 Time- 1.5 Hours		Sem - II
Instructions - 1) Answer all questions. 2) Assume suitable data.		Max.Marks- 50
Q.1	Solve any four	20 marks
	a. What are window registry? List the main registry hives in windows and the tool used to edit them.	5 marks
	b. Discuss the importance of special files in malware analysis.	5 marks
	c. List five different commands used in volatility with their syntax and purpose.	5 marks
	d. Explain difference between kernel mode and user mode in detail.	5 marks
	e. Explain how fake internet works and how it is used in malware analysis.	5 marks
Q.2	Attempt all	15 marks
	a. Convert the following program in assembly language to C language. <pre> mov dword ptr [ebp-8], 1 mov dword ptr [ebp-4], 0 loc_401014: cmp dword ptr [ebp-4], 4 jge short loc_40102E mov eax, [ebp-8] add eax, [ebp-4] mov [ebp-8], eax mov ecx, [ebp-4] add ecx, 1 mov [ebp-4], ecx jmp short loc_401014 loc_40102E: </pre> <i>Handwritten notes:</i> $if(a \geq 4)$ $b = b + a$ $a = a + 1$	5 marks
	b. Explain how does fileless malware operate using window APIs.	5 marks
	c. List Arithmetic, bitwise, conditional and unconditional with syntax and example for each.	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
	I. Explain what is function prologue and function epilogue using stack with supporting assembly code.	8 marks
	OR	
	II. Write assembly program for the following code <pre> if (x == 0) { x = 5; } </pre>	8 marks

	<pre> else if (x == 1) { x = 6; } else { x = 7; } </pre>	
Q.3 b	Attempt any one	7 marks
	I. Explain flag register and explain each flag in detail with suitable example. Discuss when are they enabled and disabled.	7 marks
	OR	
	II. List any five tools used for dynamic analysis of malware and explain how these tools are used for malware analysis.	7 marks

*** All the best***